

Requirements Checklist

1. Organization-Based Strategies

- What are your key business goals and how does this workload support them?
 - Are there any organizational standards or mandates (e.g., approved regions, architectures)?
 - Who are the primary stakeholders and decision-makers for this workload?
 - Are there any existing enterprise architectures or patterns this solution must align with?
 - How is success measured for this workload?
 - Are there dependencies on other internal systems or business units?
 - Are there timelines or milestones we should be aware of?
 - Are there future plans (e.g., migrations, upgrades, product launches) that could impact this workload?
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2. Security

- What are your security and privacy requirements (e.g., encryption, data residency)?
 - How do you currently manage identity and access?
 - Are there specific threat models or known risks for this workload?
 - Do you require centralized or decentralized security controls?
 - What is your approach to vulnerability management and patching?
 - How do you handle secrets management (API keys, credentials)?
 - Do you conduct regular security assessments or penetration testing?
 - Are specific AWS services (e.g., KMS, IAM Access Analyzer, GuardDuty) already in use?
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3. Storage Needs

- What types of data will you store (structured, unstructured, media, logs)?

Requirements Checklist

- What are your requirements for data durability, availability, and latency?
 - How frequently is the data accessed (hot, warm, cold)?
 - Are there retention policies or archival needs?
 - Do you require object storage, block storage, or file systems—or a combination?
 - How do you back up data today?
 - Are there cross-region replication or multi-AZ storage requirements?
 - How do you plan to handle data deletion or lifecycle transitions?
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4. Monitoring and Alerting

- What operational metrics do you need to monitor (e.g., latency, error rates)?
 - What is your incident response process?
 - Who should receive alerts and how (email, SMS, Slack)?
 - Do you require centralized observability across teams?
 - Do you use existing observability tools (e.g., Datadog, Splunk, CloudWatch)?
 - What are your SLIs (Service Level Indicators) and SLOs (Objectives)?
 - Do you need custom dashboards or visualization for different teams?
 - How is alert noise or false positives currently handled?
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5. Resilience

- Do you require multi-AZ or multi-region deployment?
- How is application state handled during failover events?
- Are there critical downstream dependencies that affect availability?

Requirements Checklist

- How often do you perform resilience or failure mode testing?
 - What is your tolerance for downtime or degraded performance?
 - What failure scenarios are critical to recover from?
 - Do you need automatic failover or just manual recovery?
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6. Disaster Recovery (DR) and Business Continuity (BC)

- What are your RTO (Recovery Time Objective) and RPO (Recovery Point Objective)?
 - What DR strategies do you currently use or plan to use (e.g., pilot light, warm standby)?
 - Are there business-critical systems that must remain available at all times?
 - Do you test your DR plan regularly? If so, how often?
 - Are there specific business hours when availability is critical?
 - Do you need cold, warm, or hot standby environments?
 - Are third-party dependencies (e.g., SaaS, other cloud providers, etc.) included in DR planning?
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7. Performance

- Do you have requirements around startup time or cold starts?
- Are there known performance pain points or complaints from users?
- Do you benchmark performance against competitors or internal standards?
- How do you differentiate between acceptable vs unacceptable performance?
- What are your latency and throughput requirements?
- Are there peak usage times or performance SLAs?
- How do you currently measure performance?

Requirements Checklist

8. Scaling

- Do you expect workload usage to fluctuate (daily, seasonally, or due to events)?
- What are your scale-out/scale-in triggers (e.g., CPU, queue length)?
- Do you need horizontal, vertical, or both types of scaling?
- What happens today when you hit capacity limits?
- Do you anticipate long-term growth in users, data, or requests?
- Do you need different scaling policies per component (e.g., API vs DB)?
- Are burst traffic scenarios (marketing events, product launches, etc.) expected?

9. Cost Optimization

- What is your budget for this workload?
- Are you open to using spot instances or serverless to reduce cost?
- Do you monitor and optimize cost regularly?
- Are you using any cost allocation tags or AWS Budgets?
- Do you want to reserve capacity (Savings Plans, RIs) for predictable workloads?
- Is there a threshold at which cost becomes unacceptable?
- Who is responsible for cost governance or approval?

10. Access, Authorization, and Authentication

- Who needs access to this system, and what level of access is appropriate?
- What identity providers (IdPs) do you use (e.g., Active Directory, SSO)?
- Do you use roles or policies to enforce least privilege?

Requirements Checklist

- Are different environments (dev/test/prod) managed separately or together?
 - How is access revoked or audited?
 - Do you need temporary credentials (e.g., for CI/CD, contractors)?
 - Do you integrate with external IdPs?
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11. Connectivity

- Will the system be accessed over the internet, VPN, or AWS Direct Connect?
 - Are there latency or bandwidth requirements for connectivity?
 - Do you require private connectivity between AWS services?
 - Are there systems that require hybrid connectivity with on-premise?
 - What network segmentation or isolation is required (e.g., public/private subnets)?
 - Are VPC peering or Transit Gateway already in use?
 - Do you require low-latency inter-region connectivity?
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12. Name Resolution

- Do you use internal or external DNS?
- Are there custom domain requirements for your services?
- Do you manage DNS records centrally or by workload?
- Are there naming conventions or DNS governance policies?
- Do you require split-horizon DNS (different resolution internally vs externally)?
- Will services be exposed using AWS Route 53 or another provider?
- Are health checks or failover configurations part of your DNS usage?

Requirements Checklist

13. Compliance

- What regulatory standards must this workload comply with (e.g., GDPR, HIPAA, SOC 2)?
- Do you need audit trails, logging, and access reports?
- Who is responsible for governance and compliance verification?
- Do you require data sovereignty or residency in specific jurisdictions?
- Are there internal audits or governance boards to report to?
- How is compliance enforcement automated (e.g., AWS Config, SCPs)?
- Do you require data access logging or forensic traceability?

14. Capacity Management

- What is the expected volume of requests, storage, and compute?
- How do you forecast usage and plan for growth?
- Are there resource limits or quotas you need to account for?
- Are autoscaling policies defined per environment?
- What are your usage growth projections (quarterly or yearly)?
- Do you have any buffer or overprovisioning strategies?
- How is underutilization identified and addressed?

15. Compute Needs

- What are the compute requirements (e.g., CPU-intensive, GPU, memory)?
- Do you require containerization (ECS, EKS) or serverless (Lambda)?
- Are there specific OS or runtime dependencies?

Requirements Checklist

- Are you currently using EC2, Lambda, ECS, or EKS? Any plans to migrate?
 - Do your workloads benefit from specific instance types (e.g., compute-optimized)?
 - Are there licensing requirements (e.g., Windows, Oracle)?
 - Do you need GPU-based compute (e.g., for ML or graphics)?
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16. Data Needs and Databases

- What types of data do you manage (transactional, analytical, time series)?
 - What are your requirements for consistency, availability, and partition tolerance?
 - Are you currently using relational, NoSQL, or other types of databases?
 - Do you require multi-region database replication or failover?
 - How often is your data updated, and by how many concurrent users?
 - Do you need analytics or business intelligence integration?
 - Are there migration plans for legacy databases?
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17. Session and State Management

- Do you need to persist user sessions across services or deployments?
- How do you currently manage state (e.g., cookies, databases, caching)?
- Is session affinity or sticky sessions a requirement?
- Are user sessions expected to persist across regions?
- Will this application rely on server-side or client-side state?
- How long should session data be retained?
- Do you require shared session stores (e.g., Redis, DynamoDB)?

